

Ritvik Gupta

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EDUCATION

University of Illinois at Urbana-Champaign

Bachelor of Science in Computer Engineering

GPA: 4.000/4.000

Expected Dec 2028

Millburn High School

High Honors, SAT: 1580, National Merit Scholarship Finalist

GPA: 4.602/4.000

Sep 2021 – Jun 2025

EXPERIENCE

Illini Electric Motorsports - Formula SAE

Circuit Design Project Lead

Sep 2025 – Present

Champaign-Urbana, IL

- Leading development of main vehicle control board handling critical driver safety faults and CAN communication
- Used Altium to design vehicle control board schematics, including power, fault management, and status indicators
- Validated circuit interconnections using detailed top-level schematic to ensure functionality before PCB layouting
- Analyzed Buck converter component datasheets to determine suitable candidates for main vehicle control board
- Designed V1 board layout, routing digital and analog signals to maximize signal integrity and thermal performance
- Executed a structured test and validation plan for each circuit to verify functionality, informing V2 changes

Wolfram Research

Education Software Development Intern

Feb 2025 – Jul 2025

Remote

- Engineered full-stack web portal for High School Summer Research Program, using Wolfram Cloud and JavaScript
- Streamlined application submission, progress tracking, and program management for 100+ students and faculty
- Designed and implemented database schema and backend logic in Wolfram Language to organize daily staff tasks

Millburn Robotics

Team Captain

Sep 2021 – Jun 2025

Millburn, NJ

- Led FIRST Tech Challenge (FTC) team 23650 MillburnX to NJ state championship overall grand finalists
- Presented fully autonomous VEX AI robots at the VEX World Championship for NASA and Google reps
- Founded the Garden State Alliance community of New Jersey based FTC robotics teams with 350+ members

PROJECTS

Cellular Automata Generation of 3D Printer Infill | Wolfram Mathematica

Sep 2024 – Mar 2025

- Implemented 3D cellular automata in Wolfram Language to algorithmically generate 3D printer infill geometries
- Parameterized infill density and connectivity rules to optimize structural efficiency and print material usage
- Conducted successful test prints on a Bambu Lab printer to validate automata-generated infill structures

Parametric GVF | Wolfram Mathematica, Java, Control Theory

Jun 2024 – Jul 2024

- Implemented a bird flocking simulation in Wolfram Language demonstrating emergent behavior from simple rules
- Integrated a guiding vector field to direct flock behavior, achieving convergence and following of parametric paths
- Presented project at the Wolfram Summer Research Program; featured as a Staff Pick on Wolfram Community

Spot@MHS | Python, Boston Dynamics

Nov 2023 – Mar 2024

- Developed comprehensive educational curriculum for the Spot robot, streamlining its use as a classroom tool
- Designed and open-sourced a modular payload system for Spot to expand its capabilities in a cost-effective manner

SignSensei | Python, MediaPipe, OpenCV, Tensorflow

Sep 2023

- Built a web-app using MediaPipe hand-recognition to teach American Sign Language through interactive feedback
- Placed in the top 15% of projects at PennApps XXIV for innovative and impactful use of real-time pose estimation

TECHNICAL SKILLS

Programming Languages: Python, C, Verilog, C++, Java, JavaScript, Wolfram Language

Frameworks and Libraries: TensorFlow, PyTorch, MediaPipe, OpenCV, pytest, Flask, React, NumPy, Matplotlib

Tools: Altium Designer, Bazel, Vivado, Git, Wolfram Mathematica